

Tamil Nadu Assembly 2026

Neutral election briefing for AtliQ Media

Codebasics Resume Project Challenge #21

ECI public data · 234 constituencies · 2021 vs 2026

In 2026, Tamil Nadu's assembly became more fragmented: TVK won 108 seats, 163 of 234 constituencies changed hands, and average winning margins fell from 11.8%

The challenge

Hackathon problem statement · RPC #21

Client: AtliQ Media needs ideas for a one-hour TV special on the 2026 TN Assembly election.

Constraints (non-negotiable):

- Election Commission of India data only — no news, exit polls, or social media
- Neutral, descriptive tone — no predictions, causal claims, or party commentary
- Pick 3 of 6 research threads and connect them into one clear story

Deliverables: 5–7 min video pitch · 8–10 slide deck · public GitHub · optional dashboard

Our solution

Data → insights → editorial run-of-show → interactive briefing

1. Reproducible pipeline

- Python ETL on ECI CSVs → AC-level metrics, Sankey flows, regional tables

2. Three lead narratives (research threads addressed)

- Margin of victory — fragmentation and close races (Q6)
- Seat flips — 2021→2026 party transitions (Q2)
- Vote share & TVK — new entrant scale by region (Q3)

3. Interactive briefing dashboard

- Editorial brief, statewide KPIs, charts, constituency explorer

4. This deck + video script — explicit 60-minute show recommendation

Technology stack

Built for reproducibility and demo-ready storytelling

Data & analytics

- Python 3 · pandas · Jupyter notebook (`notebooks/01\_build\_and\_explore.ipynb`)
- Shared metrics module (`src/metrics.py`, `docs/METRIC\_DEFINITIONS.md`)

Backend

- FastAPI — REST API over `data/processed/` CSVs

Frontend

- React · TypeScript · Vite · Apache ECharts

Deliverables automation

- `scripts/build\_processed\_data.py` · `build\_presentation.py` · Playwright E2E

Deployment

- Local: uvicorn :8000 · optional ngrok for sharing

Data & methodology

How we turned ECI files into comparable 2021 vs 2026 metrics

Sources

- `tn\_2021\_results.csv`, `tn\_2026\_results.csv`, `constituency\_master.csv` (234 ACs)

Join key: assembly constituency number (`ac\_number`) — not constituency name

Definitions

- Valid votes = sum of candidate votes excluding NOTA
- Winner = highest votes among non-NOTA candidates per AC
- Margin % = $(\text{winner} - \text{runner-up}) / \text{valid votes} \times 100$
- Flip = normalized 2021 winner party $\neq$ normalized 2026 winner party

Party labels normalized before seat and flip counts (see README mapping table)

Research questions — coverage map

Hackathon brief: pick 3 of 6 · we go deep on Q6/Q2/Q3 and cover Q1/Q4/Q5 as supporting views

- Q1 Geography → flip rate Chennai Metro 93.8% vs Delta 45.5%
- Q2 Flips (LEAD) → 163/234 ACs · 69.7% of state
- Q3 Vote share (LEAD) → TVK 35.1% · DMK 2021 38.0% (statewide + by-region)
- Q4 Reserved → GEN/SC/ST flip rates, margin distribution, party × category table (2021 vs 2026)
- Q5 Turnout → state record 2026 ~85.1% · per-AC delta chart in dashboard when ECI Form-20 is loaded
- Q6 Margins (LEAD) → avg 11.8% → 7.7% · 61 winners <35% · 103 races <5pp

Plus deep electoral-science measures: ENP 3.74→4.29 · Pedersen 36.4 · Gallagher LSq 15.2→8.7

Q1 — Geographic pattern

Where flips and TVK seats cluster

234-tile mosaic

- One square = one AC · equal visual weight

Regional contrast

- Top flip rate: Chennai Metro (93.8%)
- Lowest: Delta (45.5%)

On air

- Pair with district map and 2021↔2026 toggle

2026 winning party by constituency (one square = one AC)



Q2 — Seat churn at scale (LEAD)

Where seats moved between 2021 and 2026

Scale

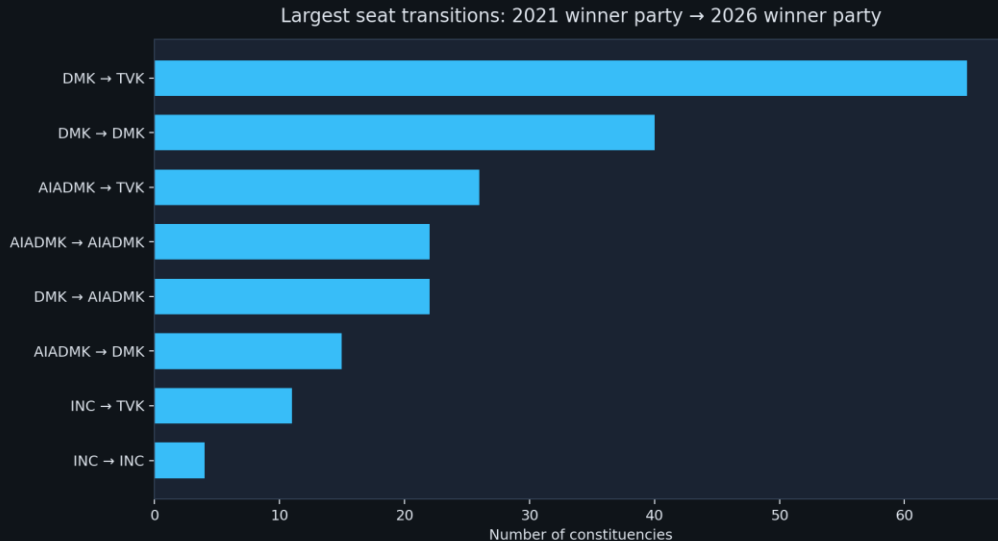
- 163 of 234 ACs changed winner
- Largest flow: DMK (2021) → TVK (2026): 65 seats

Regional pattern

- Highest flip rate: Chennai Metro (93.8%)
- Lowest: Delta (45.5%)

Neutral framing

- Say constituencies changed hands — not voters switched allegiance



Q3 — Vote share statewide (LEAD)

Where TVK landed at the state level — descriptive comparison

Statewide

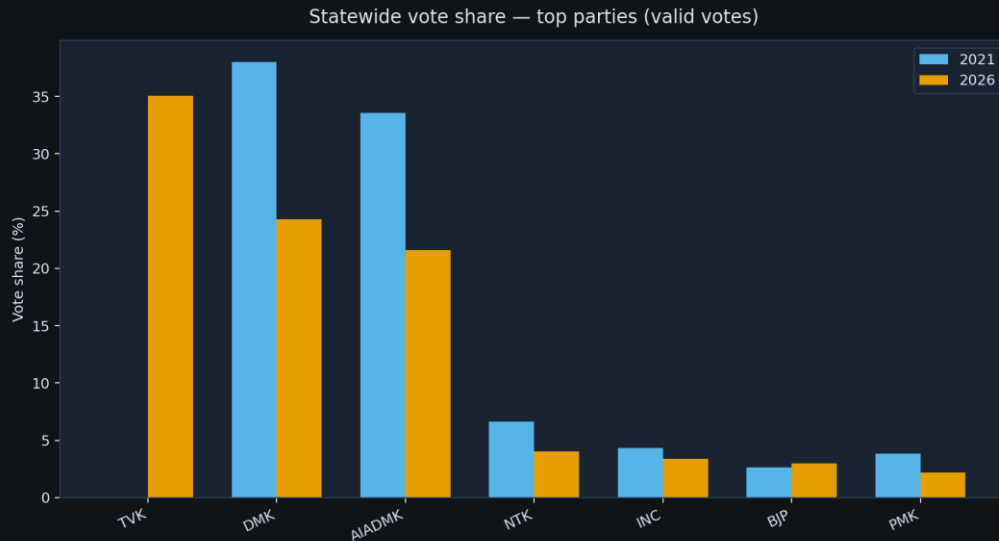
- TVK: 108 seats · ~35.1% valid vote share
- DMK 2021 baseline: 133 seats · ~38.0% share

Regional seats won (TVK)

- Chennai Metro (29), South (26), Kongu (16)

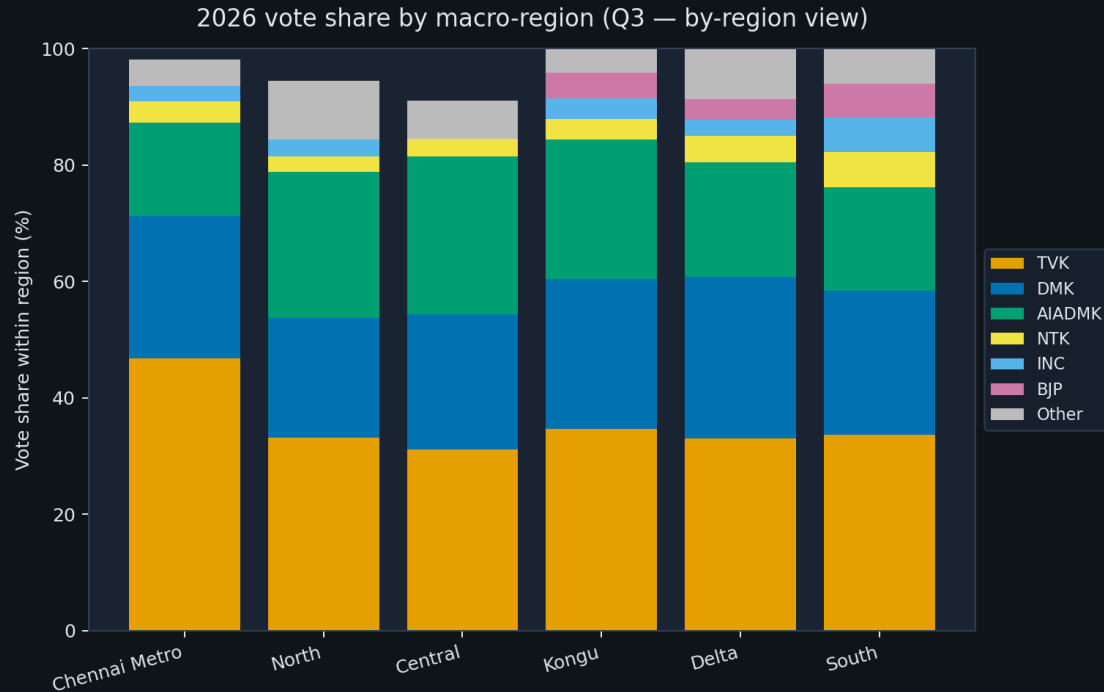
Neutral framing

- Report shares side-by-side — do not claim whose votes moved where



Q3 — Vote share by region

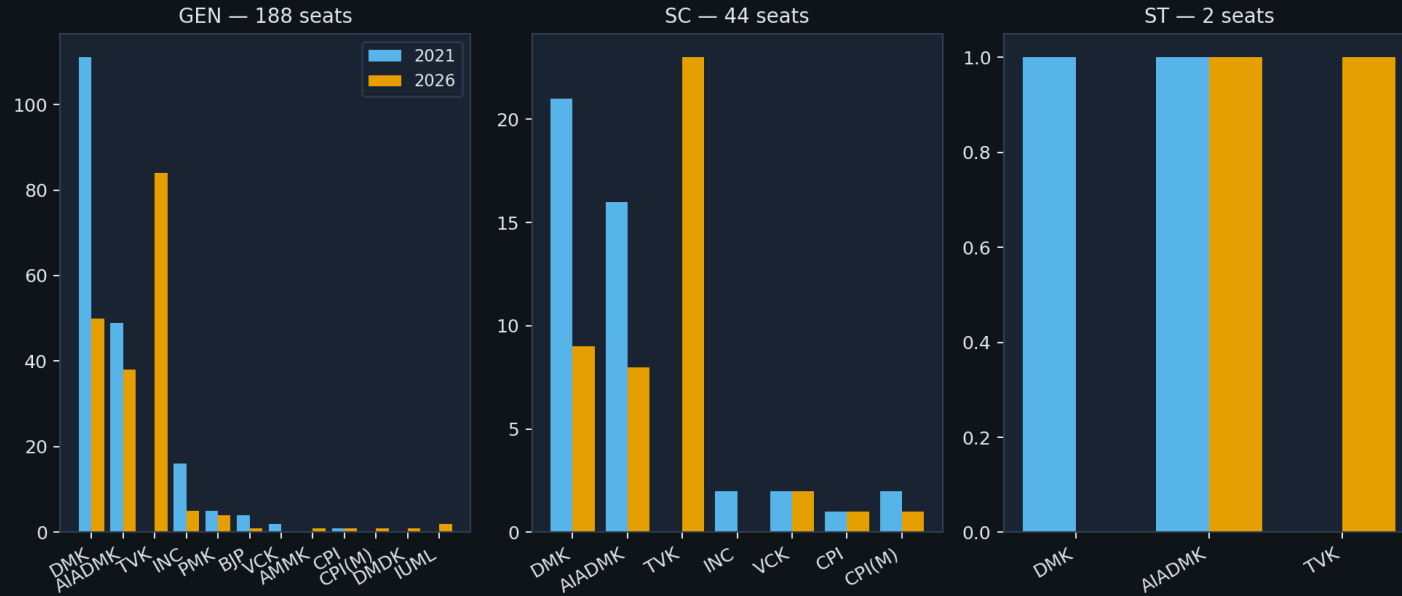
Region-level composition shows where TVK is heavier or lighter



Q4 — Reserved seats (GEN / SC / ST)

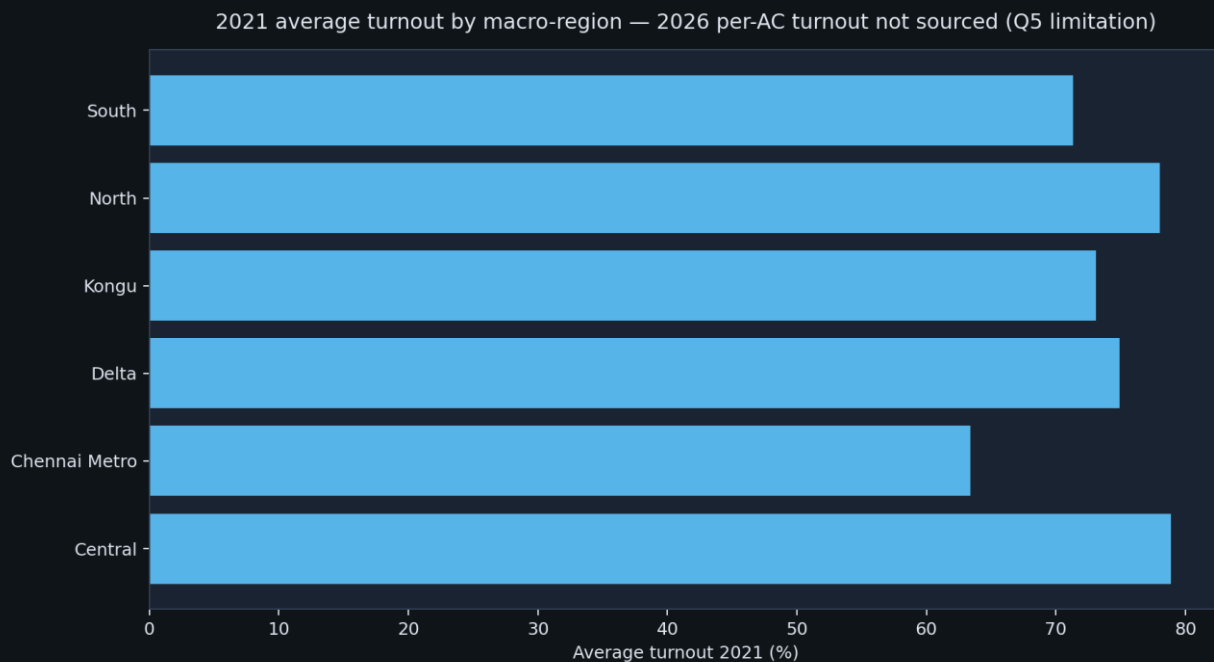
Did reserved constituencies behave differently from general seats?

Reserved-seat winners — 2021 vs 2026 (Q4)



Q5 — Turnout

Statewide record · per-AC delta requires ECI Form-20 ingestion



Q6 — Margin of victory (LEAD)

Closer races, fewer landslides

What changed

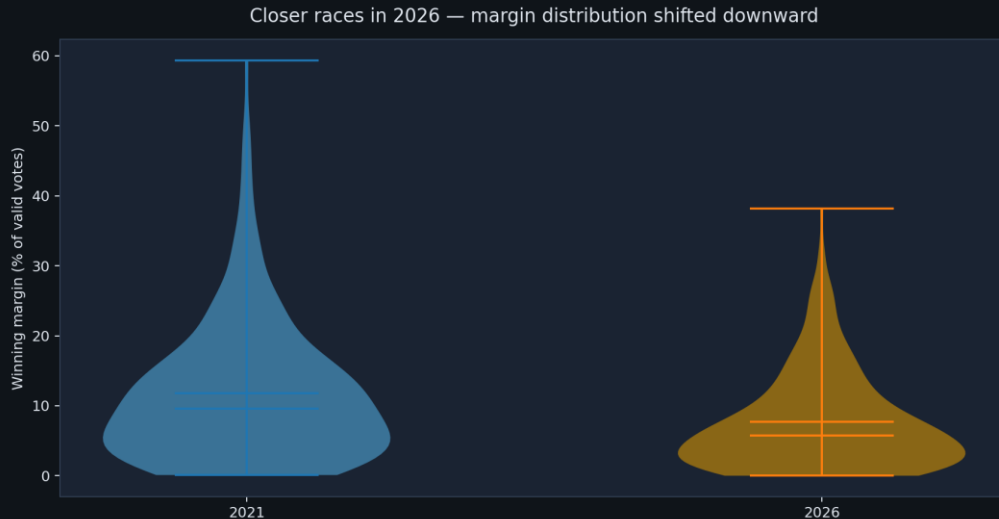
- Mean margin: 11.8% → 7.7%

What it means for TV

- 103 seats under 5-point margin
- 61 plurality winners (<35% share)

Neutral framing

- Describe distribution shift — do not explain voter motivation



Excludes NOTA from valid votes | Source: Codebasics ECI starter pack

Deep insight — electoral arithmetic

Indices that summarise the structural shift between 2021 and 2026

Fragmentation

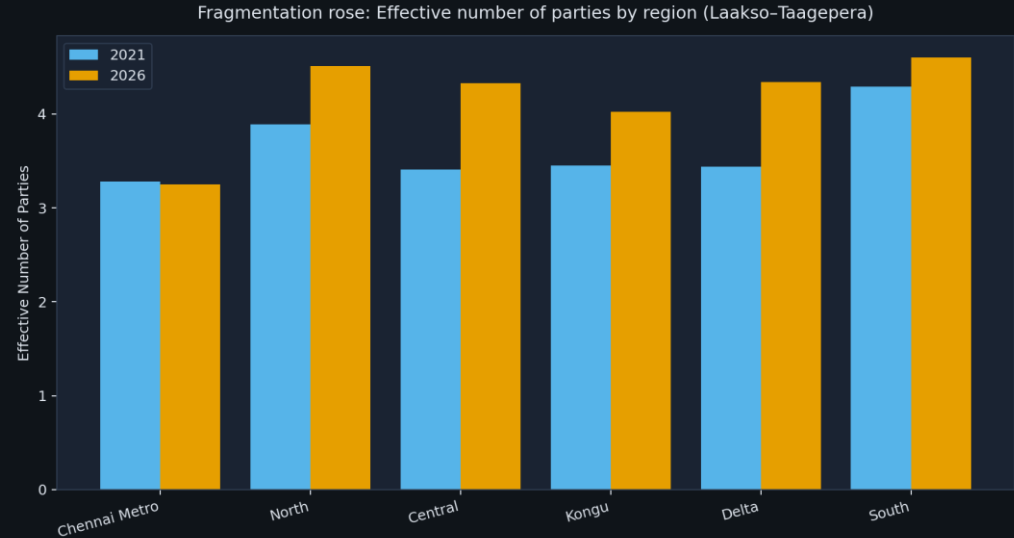
- Effective Number of Parties: 3.74 → 4.29
- Multi-cornered races (top-2 < 70%): 132 of 234

Volatility & proportionality

- Pedersen index: 36.4 — above 20 is 'high' in literature
- Gallagher LSq: 15.2 → 8.7

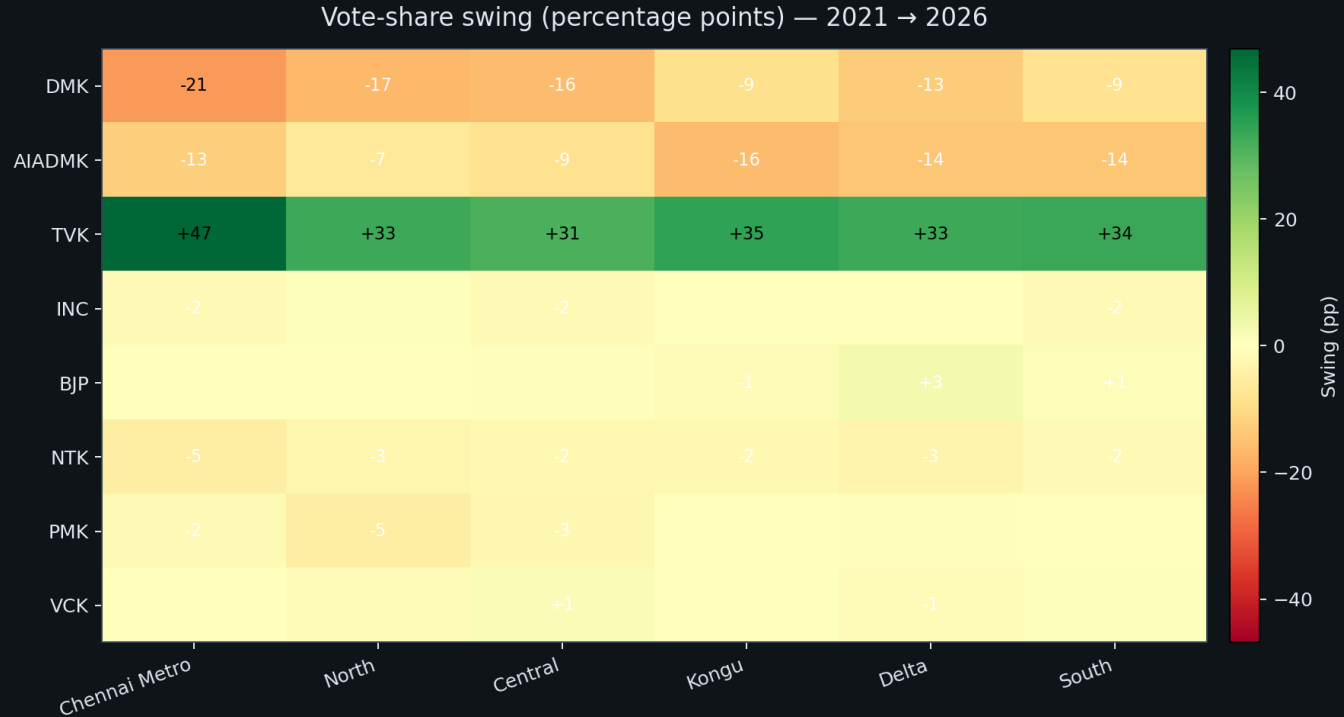
Neutral framing

- These are descriptive electoral-science measures — no causal claim



Deep insight — regional swing

Where each party gained or lost vote share between cycles



Deep insight — representation gap

First-past-the-post amplifies leading parties — by how much in 2026?

How it works

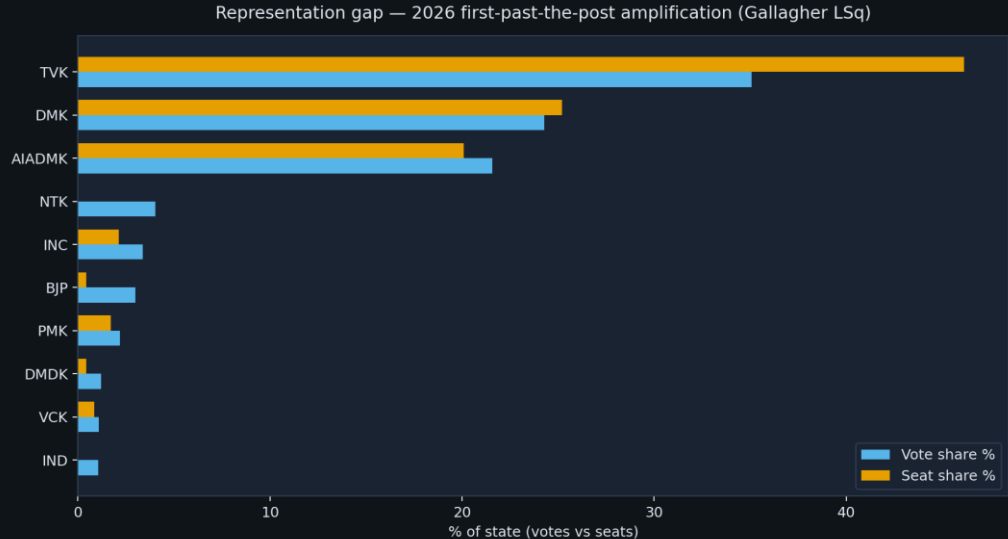
- Vote share = % of valid votes statewide
- Seat share = % of 234 ACs won
- Gap = seat share – vote share (pp)

What the data shows (2026)

- TVK gap: +11.1 pp (leading-party bonus)
- BJP gap: -2.6 pp (small parties penalised)

On air

- Use to explain why seat count alone overstates dominance



Limitations & reproducibility

Limitations

- 2026 turnout not in starter CSV — Q5 optional; 2021 turnout only in dashboard
- Constituency-level data — no booth-level or voter-age tables
- Party labels normalized — raw ECI strings differ slightly from seat counts
- Descriptive only — no forecasts or causal explanations

Reproduce this analysis

- ``python scripts/build_all_deliverables.py``
- Dashboard: ``uvicorn api.main:app --port 8000`` (after ``cd web && npm run build``)

Thank you — questions on methodology and constituency drill-down welcome